Superpixel-Level Hybrid Discriminant Analysis for Hyperspectral Image Feature Extraction

Shuzhen Zhang, *Member, IEEE*, Ting Lu, *Member, IEEE*, Wei Fu, *Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— For hyperspectral image (HSI) classification, it is a challenging problem to learn highly discriminative features, since the complex local/nonlocal spatial–spectral association is difficult to be accurately characterized. Focused on this issue, a novel superpixel-level hybrid discriminant analysis (SHDA) method is proposed, in this article. Here, the SHDA method takes advantage of superpixel's merit in characterizing spatial–spectral shape-adaptive structure and the powerful capability of discriminant analysis in enhancing class-separability to learn the feature representation. Moreover, the local/nonlocal spatial–spectral correlation information among/between superpixels is effectively excavated and fused in one framework to further improve the classification performance of features. This is achieved by first designing two specific discriminant analysis modules, i.e., superpixel-level local discriminant analysis (SLDA) and superpixel-level nonlocal discriminant analysis (SNDA). In the SLDA, adaptively weighted scatter matrices are defined to characterize the local spectral similarity within each superpixel and the discrepancy among adjacent superpixels. In the SNDA, superpixel-level graphs are built for capturing the nonlocal contextual information, where the weights of graphs are estimated based on the most similar and dissimilar superpixels. Then, the SLDA and the SNDA are effectively fused to construct the total intra/intersuperpixel scatter matrices. Finally, a joint projection transformation is obtained by solving a simple generalized eigenvalue problem. In this way, the HSI data can be projected from a high-dimensional data space into a low-dimensional feature space, where different classes of land covers can be more accurately distinguished. Experimental results on three real hyperspectral datasets indicate that the proposed SHDA method outperforms several state-of-the-art techniques.**

***Index Terms*— Classification, discriminant analysis, feature extraction, hyperspectral image (HSI), projection transformation, superpixel.**

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I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of narrow and contiguous spectral bands, which can provide subtle spectral discrepancy of different ground objects to benefit various applications in remote sensing [1], [2], [3], [4], [5]. In most of these applications, classifying each pixel in the scene is commonly required [6], [7]. However, in the condition of limited labeled samples, high spectral dimensionality may lead to the curse of dimensionality [8], which deteriorates the classification performance [7], [9], [10], [11]. Moreover, the complex spatial–spectral correlation is difficult to be characterized. To alleviate the above problems, different feature extraction methods for HSI are explored.

As a representative feature extraction approach, projection transformation projects the high-dimensional data into a reduced-dimensional space to capture the underlying characteristics of high-dimensional data. Two typical projection transformation methods are principal component analysis (PCA) [12] and linear discriminant analysis (LDA) [13]. PCA is an unsupervised approach, which is to seek an optimal linear projection to maximize the data variance. LDA is a supervised projection transformation method, which finds the projection direction that minimizes the ratio of the within-class and the between-class scatter. Subsequently, various PCA extensions [14], [15], [16] and LDA extensions [17], [18], [19] have been proposed. Besides, manifold learning methods represented by local preserving projection [20] and neighbor preserving embedding (NPE) [21], which aim to find the low-dimensional space to keep the local geometric structure of the high-dimensional data, have also been utilized for feature extraction of images. Moreover, there are some supervised graph-embedding models [22], [23], [24], [25] are developed, which try to separate the submanifold of each class from each another. Recently, to further explore the intrinsic relationship of data, some new techniques have been introduced in discriminant analysis methods. For instance, in [26] and [27], a semi-supervised learning strategy is used to learn the discriminant projection by efficiently utilizing few labeled samples and abundant unlabeled samples. In [28], the hypergraph model is employed to reveal the complex high-order relationships of HSI. In [29], multistructure properties of a sample are

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combined into a unified embedding framework, which contributes to obtaining more discriminative features. Although these adopted techniques can enhance the class separability of generated features, most of them mainly focus on the use of spectral information, which neglects the spatial structure information of samples.

To exploit both the spatial and spectral information, many projection transformations based spatial–spectral feature extraction technique have been proposed for HSI in recent decades. For example, Zhou et al. [23] developed a local pixel neighborhood preserving embedding (LPNPE), which seeks a linear projection to minimize the local pixel neighborhood preserving scatter and maximize the total scatter in the projected space. Huang et al. [30] adopted a spatial–spectral combined distance to construct the adjacency graph of NPE. Deng et al. [31] proposed a patch tensor-based multigraph embedding framework, which explores the local and nonlocal information of pixel patches to obtain the projection matrix. Huang et al. [32] put forward a spatial–spectral local discriminant projection method based on local spatial consistency and nonlocal spectral similarity. Although these spatial–spectral feature extraction methods do yield impressive results in the HSI classification, they exploit the spatial–spectral information within a fixed-size spatial window. The main issue related to the above methods lies in that the spatial–spectral information extracted from subregions with fixed-size and fixed-shape can not provide sufficient adaptive structural information [33].

Furthermore, superpixel segmentation which can divide an image into nonoverlapping patches of structure-adaptive size and shape has been developed to be a promising tool for feature extraction of HSI [33], [34], [35], [36], [37], [38]. For example, in [35] and [36], first-order or second-order statistical features of the superpixel are extracted for HSI classification. In [37] and [38], PCA is applied on each superpixel to obtain the unsupervised features. In these feature extraction methods, the superpixel is commonly regarded as an independent unit. The correlation among different superpixels is neglected, which actually contributes to improve the discernibility of the extracted features. Recently, some research studies are trying to explore the associated information among different superpixels in feature extraction [39], [40], [41], [42], [43]. For instance, in [39] and [40], neighboring superpixels are utilized to generate the weighted average feature. In [41], [42], and [43], any two superpixels are considered as two pixel sets with different class labels, which is applied in superpixel-based discriminant analysis. However, the existing superpixel-based feature extraction methods for HSI classification mainly focused on local pixelwise or superpixelwise correlation, neglecting the extraction and fusion of both the local and nonlocal spatial–spectral correlation information between and among different superpixels.

Based on the above analysis, we propose a novel superpixel-level hybrid discriminant analysis (SHDA) for the feature extraction of HSI. In this method, discriminant analysis is first conducted from superpixel-level local and nonlocal viewpoints, respectively. On one hand, spectral-similarity information within each superpixel and diversity information among adjacent superpixels are explored in the superpixel-level local discriminant analysis (SLDA). On the other hand, relationship information among different superpixels is excavated in the superpixel-level nonlocal discriminant analysis (SNDA). After that, the proposed SHDA framework effectively fuses the SLDA and the SNDA to learn a projection matrix and transforms the HSI from the image space into a discriminative feature space. The major contributions of this article are summarized as follows.

1) To learn better feature representation for HSI classification, a novel superpixel-oriented discriminant analysis method is proposed in this article. This method can take advantage of superpixel's superiority in characterizing spatial–spectral shape-adaptive structure and the powerful learning capability of discriminant analysis in enhancing class-separability.
2) To fully exploit the rich local/nonlocal spatial–spectral information in HSI, two separate discriminant analysis modules are designed to measure the complex correlation within and among superpixels. This is achieved by the introduced SLDA module based on the calculation of local weighted scatter matrices and SNDA module via the construction of nonlocal graphs.
3) Aiming at effectively integrating the SLDA/SNDA modules in one framework, a joint projection transformation is constructed to transform the issue of feature extraction to solve a simple generalized eigenvalue problem. In this way, the original HSI can be projected from the high-dimensional data space into a low-dimensional feature space, where the capability in separating different classes can be greatly boosted.

The rest of this article is structured as follows. Section II briefly introduces the related works, including the LDA method and the local discriminant embedding (LDE) [22] method. In Section III, the proposed SHDA method is described in detail. In Section IV, the experimental results and analysis are provided. Finally, some conclusions are given in Section V.

II. Related Works

A. Linear Discriminant Analysis

The LDA is a representative supervised projection transformation method. In LDA, the intraclass compactness and the interclass separability are, respectively, described by the intraclass scatter matrix $\mathbf{S}_w$ and the interclass scatter matrix $\mathbf{S}_b$, which are defined as follows:

$$\mathbf{S}_w = \sum_{k=1}^{c} \sum_{i=1}^{n_k} \left(\mathbf{x}_k^i - \boldsymbol{\phi}_k\right)\left(\mathbf{x}_k^i - \boldsymbol{\phi}_k\right)^T \tag{1}$$

$$\mathbf{S}_b = \sum_{k=1}^{c} n_k \left(\boldsymbol{\phi}_k - \boldsymbol{\phi}\right)\left(\boldsymbol{\phi}_k - \boldsymbol{\phi}\right)^T \tag{2}$$

where c is the number of classes, n_k is the number of samples in the kth class, $\mathbf{x}_k^i$ is the ith sample from the kth class, $\boldsymbol{\phi}_k$ is the mean of the kth class, and $\boldsymbol{\phi}$ is the mean of all the samples. To enhance the compactness within the same class and the

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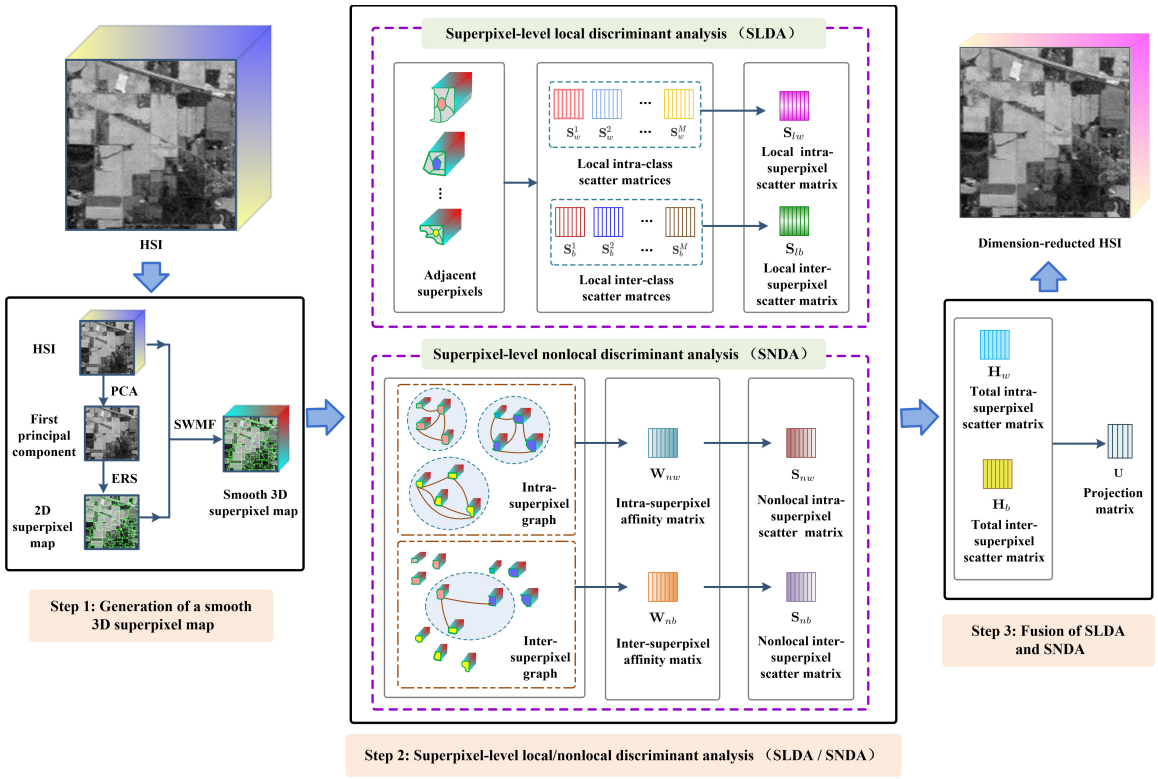


Fig. 1. Flowchart of the proposed SHDA method.

separability among different classes, the optimization problem for LDA is given as follows:

$$\arg\min_{\mathbf{U}} \frac{tr\left(\mathbf{U}^T\mathbf{S}_w\mathbf{U}\right)}{tr\left(\mathbf{U}^T\mathbf{S}_b\mathbf{U}\right)} \tag{3}$$

where $tr(\cdot)$ represents the trace operator. The optimal projection matrix $\mathbf{U}$ can be obtained by solving the generalized eigenvalue problem.

B. Local Discriminant Embedding

The LDE is a graph embedding framework to learn the transformation projection. The LDE method consists of three steps. Specifically, to characterize the geometrical structure of the data manifold, a intraclass graph $\mathbf{G}_w$ and an interclass graph $\mathbf{G}_b$ are first constructed. Given a dataset $\boldsymbol{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n]$, where n is the number of data point. To construct $\mathbf{G}_w$, for each pair of points $\mathbf{x}_i$ and $\mathbf{x}_j$ from the same class, an edge is added between them if $\mathbf{x}_i$ is the δ_w-nearest neighbors (NNs) of $\mathbf{x}_j$. To construct $\mathbf{G}_b$, considering each pair $\mathbf{x}_i$ and $\mathbf{x}_j$ from the different classes, an edge is added between them if $\mathbf{x}_i$ is the δ_b-NNs of $\mathbf{x}_j$. Second, the affinity weight matrix $\mathbf{W}_w$ for the intraclass graph $\mathbf{G}_w$ and the affinity weight matrix $\mathbf{W}_b$ for the interclass graph $\mathbf{G}_b$ are, respectively, established. All elements of the two affinity weight matrices are computed by using a heat kernel function, which is defined as

$$W_{ij} = \exp\left(-\left\|\mathbf{x}_i - \mathbf{x}_j\right\|_2^2 / t\right) \tag{4}$$

where t is the heat kernel parameter. The W_{ij} represents the weight of the edge between $\mathbf{x}_i$ and $\mathbf{x}_j$. Finally, the optimal projection matrix $\mathbf{U}$ is obtained by solving the following optimization problem:

$$\arg\min_{\mathbf{U}} \frac{tr\left(\mathbf{U}^T\mathbf{X}\mathbf{L}_w\mathbf{X}^T\mathbf{U}\right)}{tr\left(\mathbf{U}^T\mathbf{X}\mathbf{L}_b\mathbf{X}^T\mathbf{U}\right)} \tag{5}$$

where $\mathbf{L}_w = \mathbf{D}_w - \mathbf{W}_w$ is the intraclass graph Laplacian matrix and $\mathbf{L}_b = \mathbf{D}_b - \mathbf{W}_b$ is the interclass graph Laplacian matrix, $\mathbf{D}_w$ and $\mathbf{D}_b$ are diagonal matrices whose entries are, respectively, column sums of $\mathbf{W}_w$ and $\mathbf{W}_b$.

III. Proposed Method

In this article, we propose a novel SHDA method, which mainly consists of the following three parts: 1) generation of 3-D superpixel map; 2) SLDA/SNDA; and 3) fusion of SLDA and SNDA. The flowchart of the proposed SHDA method is illustrated in Fig. 1 and the details of the method are given as follows.

A. Generation of 3-D Superpixel Map

In HSI, neighboring pixels always present similar spectral characteristics. To effectively capture the spatial correlation and spectral similarity, as shown in Fig. 1, a smooth 3-D superpixel map is first generated, which contains multiple shape-adaptive smooth 3-D superpixels. Specifically, the PCA approach is first conducted on the HSI to obtain the first principal component, which contains the major information of HSI. Then, the entropy rate superpixel (ERS) segmentation [44] technique, which is a graph-based superpixel segmentation method, is applied on the first principal component to produce

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a 2-D superpixel map, which contains M nonoverlapping shape-adaptive superpixels. Since the ERS is not the original work of this article, we will not give too much illustration of it. Refer [44] to get more details of this oversegmentation technique. After that, the 2-D superpixel map combines the original HSI to generate a 3-D image marked by superpixels. For each superpixel of the 3-D image, a superpixel-based weighted mean filter (SWMF) [38] is applied to reduce noise and the corresponding smooth 3-D superpixel is achieved, which is used to construct the smooth 3-D superpixel map.

For the superpixel segmentation, the superpixel number M must be determined in advance. Generally, the more complex of the spatial construction of an image is, the more number of superpixels should be set up. At the same time, the image edge can reflect the spatial complexity of an image. Therefore, the superpixel number is computed as follows [36]:

$$M = \mu \times N_{\text{edge}} \tag{6}$$

where N_{edge} is the number of the nonzero pixel in the edge image, which is obtained by conducting Sobel filter [45] on the first principal component of HSI. μ is a proportionality coefficient.

The process of SWMF is as follows. For one pixel $\mathbf{x}_i$ in an original 3-D superpixel block, we denote its p nearest spatial neighbors in the block as $\boldsymbol{X}_i^s = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_p]$. The SWMF for the pixel $\mathbf{x}_i$ is defined as

$$\mathbf{x}_i^* = \frac{\sum_{j=1}^{p} \upsilon_{ij}\mathbf{x}_j}{\sum_{j=1}^{p} \upsilon_{ij}} \tag{7}$$

where $\upsilon_{ij} = \exp(- \parallel \mathbf{x}_i - \mathbf{x}_j \parallel_2^2 / 2\beta_i^2)$ is the weight to measure spectral similarity between pixel $\mathbf{x}_i$ and its neighboring pixel $\mathbf{x}_j$. $\beta_i = (1/p)\sum_{j=1}^{p} \parallel \mathbf{x}_i - \mathbf{x}_j \parallel_2^2$ is a kernel parameter.

In the SWMF, the number of NN pixels p is related to the area of superpixel and the spatial resolution of HSI I_{res}. In general, the larger the average area of 2-D superpixel is, the larger the value of p should be. Moreover, higher spatial resolution (i.e., smaller value of I_{res}) means more detailed spatial structure. In this case, a smaller size of the neighborhood is more appropriate to extract highly similar pixels. Therefore, the value of p should be monotonically increased with the growth of the value of I_{res}. Based on the above analyses, p is calculated by the following formula:

$$p = \sqrt[8]{I_{\text{res}}} \times \frac{I_{\text{row}} \times I_{\text{col}}}{M} - \varepsilon \tag{8}$$

where $I_{\text{row}} \times I_{\text{col}}$ represents the spatial area of HSI. ε is a correction term, which is used to fine-tune the value of p. I_{res} is measured in meters and the 8th root of I_{res} in the formula (8) is the empirical set.

B. Superpixel-Level Local/Nonlocal Discriminant Analysis

To fully and reasonably characterize the spatial–spectral information based superpixel, in this section, we, respectively, conduct the discriminant analysis from the superpixel-level local and nonlocal viewpoints.

1) Superpixel-Level Local Discriminant Analysis: In SLDA, the key is to construct two superpixel-based local scatter matrices for one superpixel and its adjacent superpixels. The process of SLDA is illustrated in Fig. 1 and a detailed description is given below.

First, for one superpixel, the local interclass scatter matrix is computed. Specifically, for the mth superpixel of the HSI, we denote $\boldsymbol{\phi}_m^s$ as the mean pixel of the mth superpixel. The mean pixels of its adjacent superpixels are denoted as $\boldsymbol{\phi}_{m,j}^s$ $(j = 1, 2, \ldots, J)$, where J is the number of adjacent superpixels. Since pixels within one superpixel can be regarded as samples from the same class, the mean pixel of one superpixel can be used as the representative feature of the superpixel. Here, we calculate the cosine similarity $C_{m,j}$ between $\boldsymbol{\phi}_m^s$ and $\boldsymbol{\phi}_{m,j}^s$, which is used to measure the similarity between the mth superpixel and its jth adjacent superpixel. When $C_{m,j}$ is less than a given threshold θ, pixels in the two superpixels can be supposed to come from two different classes. Then, the jth adjacent superpixel should be considered in the calculation of the local interclass scatter matrix. Therefore, for the mth superpixel, the local interclass scatter matrix $\mathbf{S}_b^m$ can be defined as follows:

$$\mathbf{S}_b^m = \sum_{j=1}^{J} P_{m,j}\left(\boldsymbol{\phi}_{m,j}^s - \boldsymbol{\phi}_m^s\right)\left(\boldsymbol{\phi}_{m,j}^s - \boldsymbol{\phi}_m^s\right)^T \alpha_j \tag{9}$$

where $P_{m,j}$ is the number of pixels in the jth adjacent superpixel. α_j represents the similarity weight between the mth superpixel and its jth adjacent superpixel, which is defined as

$$\alpha_j = \begin{cases} \frac{\exp\left(-\|\boldsymbol{\phi}_{m,j}^s - \boldsymbol{\phi}_m^s\|_2^2/t\right)}{\sum_j \exp\left(-\|\boldsymbol{\phi}_{m,j}^s - \boldsymbol{\phi}_m^s\|_2^2/t\right)}, & C_{m,j} < \theta \\ 0, & C_{m,j} \geq \theta. \end{cases} \tag{10}$$

In formula (10), t is the heat kernel parameter, which is empirically set to 0.1. $C_{m,j} = ((\boldsymbol{\phi}_m^s)^T \cdot \boldsymbol{\phi}_{m,j}^s / \parallel \boldsymbol{\phi}_m^s \parallel_2 \cdot \parallel \boldsymbol{\phi}_{m,j}^s \parallel_2)$. The threshold θ is experientially set to 0.85.

Second, the local intraclass scatter matrix of each superpixel is computed. In specific, we denote $\mathbf{x}_m^{(i)}$ $(i = 1, 2, \ldots, P_m)$ as the ith pixel of the mth superpixel, where P_m is the number of pixel in the mth superpixel. For the mth superpixel, the local intraclass scatter matrix $\mathbf{S}_w^m$ is defined as

$$\mathbf{S}_w^m = \sum_{i=1}^{P_m} N_m\left(\mathbf{x}_m^{(i)} - \boldsymbol{\phi}_m^s\right)\left(\mathbf{x}_m^{(i)} - \boldsymbol{\phi}_m^s\right)^T \beta_i \tag{11}$$

where β_i represents the similarity weight between the pixel $\mathbf{x}_m^{(i)}$ and the mean pixel of the mth superpixel, which is defined as

$$\beta_i = \frac{\exp\left(-\left\|\mathbf{x}_m^{(i)} - \boldsymbol{\phi}_m^s\right\|_2^2/t\right)}{\sum_{i=1}^{P_m} \exp\left(-\left\|\mathbf{x}_m^{(i)} - \boldsymbol{\phi}_m^s\right\|_2^2/t\right)}. \tag{12}$$

$(N_m - 1)$ is the number of adjacent superpixels that satisfy $C_{m,j} < \theta$, i.e., pixels in these adjacent superpixels and pixels in the mth superpixel belong to different classes.

After defining two local scatter matrices for each superpixel, the local intrasuperpixel scatter matrix $\mathbf{S}_{lw}$ and the local

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intersuperpixel scatter matrix $\mathbf{S}_{lb}$ for the HSI can be obtained by the following formulas:

$$\mathbf{S}_{lw} = \sum_{m=1}^{M} \mathbf{S}_w^m \qquad \mathbf{S}_{lb} = \sum_{m=1}^{M} \mathbf{S}_b^m. \tag{13}$$

Similar to LDA, the optimization function of the proposed SLDA is defined as

$$\arg\min_{\mathbf{U}} \frac{tr\left(\mathbf{U}^T \mathbf{S}_{lw} \mathbf{U}\right)}{tr\left(\mathbf{U}^T \mathbf{S}_{lb} \mathbf{U}\right)}. \tag{14}$$

The optimal projection can be obtained by the minimum eigenvalue solution to the generalized eigenvalue problem.

2) Superpixel-Level Nonlocal Discriminant Analysis: Since pixels belonging to the same superpixel have strong spectral similarity, it is reasonable to regard each superpixel as a single entity. Based on the assumption, the SNDA method is proposed, in which the relationship among different superpixels is explored in the whole HSI. Specifically, let $\Phi = [\boldsymbol{\phi}_1^s, \boldsymbol{\phi}_2^s, \ldots, \boldsymbol{\phi}_m^s, \ldots, \boldsymbol{\phi}_M^s]$ represent the matrix consisting of mean pixels of all superpixels. Then, cosine similarity C_d of mean pixels of any two superpixels is calculated, which is used to measure the similarity between the two superpixels. According to C_d, for each superpixel, the most δ_w^s similar superpixels and the least δ_b^s similar superpixels are obtained, which are then used to construct the intrasuperpixel graph and the intersuperpixel graph, respectively. Finally, similar to the LDE method, the projection matrix for feature extraction can be obtained.

The mathematical description of the SNDA method is as follows. In specific, for the mth superpixel, let $\boldsymbol{\phi}_{w_j}^{s_w}$ $(w_j = 1, 2, \ldots, \delta_w^s)$ and $\boldsymbol{\phi}_{b_j}^{s_b}$ $(b_j = 1, 2, \ldots, \delta_b^s)$ be mean pixels of the most δ_w^s similar superpixels and mean pixels of the least δ_b^s similar superpixels, respectively. Then, the affinity weights $W_m^{w_j}$ between the mth superpixel and its most δ_w^s similar superpixels are defined as

$$W_m^{w_j} = \exp\left(- \parallel \boldsymbol{\phi}_m^s - \boldsymbol{\phi}_{w_j}^{s_w} \parallel_2^2 / t\right), \quad \left(w_j = 1, 2, \ldots, \delta_w^s\right). \tag{15}$$

Similar, for the least δ_b^s similar superpixels of the mth superpixel, the affinity weights can be defined as

$$W_m^{b_j} = \exp\left(- \parallel \boldsymbol{\phi}_m^s - \boldsymbol{\phi}_{b_j}^{s_b} \parallel_2^2 / t\right), \quad \left(b_j = 1, 2, \ldots, \delta_b^s\right). \tag{16}$$

For all superpixels, the two affinity weights (i.e., $W_m^{w_j}$ and $W_m^{b_j}$, $m = 1, 2, \ldots, M$) are calculated. Then, the corresponding affinity matrices $\mathbf{W}_{nw}$ and $\mathbf{W}_{nb}$ are, respectively, constructed for the intrasuperpixel graph and the intersuperpixel graph. Similar to the LDE method, the intrasuperpixel Laplacian matrix and intersuperpixel Laplacian matrix, i.e., $\mathbf{L}_{nw}$ and $\mathbf{L}_{nb}$, can be defined as

$$\mathbf{L}_{nw} = \mathbf{D}_{nw} - \mathbf{W}_{nw} \qquad \mathbf{L}_{nb} = \mathbf{D}_{nb} - \mathbf{W}_{nb} \tag{17}$$

where $\mathbf{D}_{nw}$ and $\mathbf{D}_{nb}$ are diagonal matrices whose entries are, respectively, column sums of $\mathbf{W}_{nw}$ and $\mathbf{W}_{nb}$. Finally, we extended the LDE to the regularized LDE (RLDE) [23], and the optimal projection $\mathbf{U}$ can be obtained by solving the following function:

$$\arg\min_{\mathbf{U}} \frac{tr\left(\mathbf{U}^T \mathbf{S}_{nw} \mathbf{U}\right)}{tr\left(\mathbf{U}^T \mathbf{S}_{nb} \mathbf{U}\right)} \tag{18}$$

where $\mathbf{S}_{nw}$ and $\mathbf{S}_{nb}$ are named as nonlocal intrasuperpixel scatter matrix and nonlocal intersuperpixel scatter matrix, respectively. They are defined as follows:

$$\mathbf{S}_{nw} = (1 - \eta)\Phi \mathbf{L}_{nw} \Phi^T + \eta \mathrm{diag}\left(\Phi \mathbf{L}_{nw} \Phi^T\right) \tag{19}$$

$$\mathbf{S}_{nb} = (1 - \eta)\Phi \mathbf{L}_{nb} \Phi^T + \eta \Phi \Phi^T. \tag{20}$$

The second term of the formula $\mathbf{S}_{nw}$ is a diagonal regularization term, which aims to improve the stability of the solution of (18). The second term of the formula $\mathbf{S}_{nb}$ is also a regularization term, which is utilized to preserve the maximal superpixel-level data variance. $0 \leq \eta \leq 1$ is a regularization parameter and $\mathrm{diag}(\mathbf{A})$ represents the diagonal part of matrix $\mathbf{A}$.

Note that, for simplicity, the number of intrasuperpixel and intersuperpixel neighbors in SNDA, i.e., δ_w^s and δ_b^s, will be set to the same value. We empirically set $\delta_w^s = \delta_b^s = 5$.

C. Fusion of SLDA and SNDA

On one hand, SLDA uses spatial–spectral information within the superpixel and differential information among adjacent superpixels to learn the discriminant projection, which aims to preserve the superpixel-level local neighborhood structure. On the other hand, SNDA from the nonlocal perspective learns the discriminant projection to preserve the superpixel-level nonlocal geometric structure. Obviously, the information used in the two analyses is complementary to each other. Therefore, as shown in Fig. 1, the proposed SHDA framework fuses SLDA and SNDA to learn the discriminant projection.

In SHDA, the total intrasuperpixel scatter matrix $\mathbf{H}_w$ and the total intersuperpixel scatter matrix $\mathbf{H}_b$ are defined as follows:

$$\mathbf{H}_w = \gamma \cdot \mathbf{S}_{lw} + (1 - \gamma) \cdot \mathbf{S}_{nw} \tag{21}$$

$$\mathbf{H}_b = \gamma \cdot \mathbf{S}_{lb} + (1 - \gamma) \cdot \mathbf{S}_{nb} \tag{22}$$

where γ is a tradeoff parameter with $0 \leq \gamma \leq 1$. In this way, the SLDA and SNDA are integrated. The optimization function of SHDA is defined as

$$\arg\min_{\mathbf{U}} \frac{tr\left(\mathbf{U}^T \mathbf{H}_w \mathbf{U}\right)}{tr\left(\mathbf{U}^T \mathbf{H}_b \mathbf{U}\right)}. \tag{23}$$

The projection matrix $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_d]$ can be effectively constructed by the d eigenvectors corresponding to the d smallest nonzero eigenvalues in the following eigenvalue problem:

$$\mathbf{H}_w \mathbf{u}_i = \lambda_i \mathbf{H}_b \mathbf{u}_i \tag{24}$$

where λ_i is the ith minimum eigenvalue.

After obtaining the projection matrix $\mathbf{U}$, for one pixel in HSI, the final spectral–spatial discriminant feature $\mathbf{y}$ can be calculated as

$$\mathbf{y} = \mathbf{U}^T \mathbf{x}^*. \tag{25}$$

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TABLE I
NUMBERS OF LABELED SAMPLES OF DIFFERENT CLASSES IN THE THREE TEST DATASETS

Indian Pines			Pavia University			Salinas		
Class	Name	Number	Class	Name	Number	Class	Name	Number
1	Alfalfa	46	1	Asphalt	6631	1	Weeds_1	2009
2	Corn-no till	1428	2	Meadows	18649	2	Weed_2	3726
3	Corn-min till	830	3	Gravel	2099	3	Fallow	1976
4	Corn	237	4	Trees	3064	4	Fallow plow	1394
5	Grass/pasture	483	5	Metal sheets	1345	5	Fallow smooth	2678
6	Grass/tree	730	6	Bare soil	5029	6	Stubble	3959
7	Grass/pasture-mowed	28	7	Bitumen	1330	7	Celery	3579
8	Hay-windrowed	478	8	Bricks	3682	8	Grapes	11271
9	Oats	20	9	Shadows	947	9	Soil	6203
10	Soybean-no till	972				10	Corn	3278
11	Soybean-min till	2455				11	Lettuce 4 wk	1068
12	Soybean-clean till	593				12	Lettuce 5 wk	1927
13	Wheat	205				13	Lettuce 6 wk	916
14	Woods	1265				14	Lettuce 7 wk	1070
15	Buildings-grass-trees-drives	386				15	Vineyard untrained	7268
16	Stone-steel towers	93				16	Vineyard trellis	1807
	Total	10249		Total	42776		Total	54129

Then, the high-dimensional HSI is projected into a low-dimensional feature space, in which the class-discriminative ability is significantly reinforced.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

In this section, we first introduce three real hyperspectral datasets used in our experiments. Second, the parameters of all compared methods and the evaluation metrics of the classification performance are given. Third, the impact of each parameter in the proposed SHDA method is discussed. Fourth, the module effectiveness in the proposed SHDA framework is evaluated. Finally, the comparison results with both classical and state-of-the-art feature extraction approaches are presented.

A. Datasets

1) Indian Pines: The Indian Pines image covers the agricultural Indian Pines test site in northwestern Indiana and was collected by the AVIRIS sensor. The dataset contains 200 spectral bands after discarding 20 water absorption bands. The spectral coverage ranges from 0.4 to 2.5 μm. The spatial size is 145 $\times$ 145 with a spatial resolution of 20 m per pixel. The dataset contains 16 classes and detailed information about labeled samples is listed in Table I.

2) Pavia University: The Pavia University image was collected by the ROSIS-03 sensor over an urban area surrounding the University of Pavia, Italy. It contains of 610 $\times$ 340 pixels with a spatial resolution of 1.3 m, and 103 spectral bands covering the spectral range from 0.43 to 0.86 μm. Nine classes are contained in the dataset and the detailed information of labeled samples is listed in Table I.

3) Salinas: The Salinas image was captured by the AVIRIS sensor over the area of Salinas Valley, CA, USA. The dataset contains 204 spectral bands after 20 water absorption spectral bands are being removed. The spatial size is 512 $\times$ 217 and the spatial resolution is 3.7 m per pixel. The dataset contains 16 classes and detailed information about labeled samples is listed in Table I.

B. Experimental Setup

To demonstrate the performance of the proposed SHDA[1] method, various feature extractions methods, including LDA [13], local Fisher discriminant analysis (LFDA) [19], RLDE [23], LPNPE [23], superpixelwise PCA (SuperPCA) [37], superpixel-level unsupervised LDA (SuULDA) [41], superpixel-based spatial–spectral dimension reduction (SSDR) [42], local covariance matrix representation (LCMR) [46] and superpixel-based Brownian descriptor (SBD) [36], are used for comparison. Considering that the proposed method is based on projection transformation, the majority of the compared methods (except for LCMR and SBD) are projection transformation ones. Among these methods, LDA, LFDA, RLDE, LPNPE, SSDR, LCMR, and SBD are supervised ones, while SuperPCA, SuULDA, and SHDA are unsupervised ones. In specific, LDA, LFDA, and RLDE learn the projection features by analyzing the spectral-domain similarity and diversity information of samples. LPNPE and LCMR extract spectral–spatial features by exploiting the spectral–spatial information within a fixed-window neighborhood. In SuperPCA, SuULDA, SSDR, SBD, and SHDA, the superpixel segmentation technique is utilized to obtain the adaptive-structure information for feature extraction. In SBD, the local feature is extracted by combing superpixel and the spectral-spatial correlation information. In SuperPCA, PCA is applied on each superpixel to realize the unsupervised dimension reduction. In SuULDA, the spectral-similarity information within the superpixel is utilized for the discriminant analysis. In SSDR, both the spectral-similarity information within the superpixel and the label information of training samples are integrated to learn the discriminant projections.

For the superpixel-level projection transformation-based methods, i.e., SuperPCA, SuULDA, and SSDR, the SWMF is first performed on the HSI to enhance image quality, and the parameter p in the SWMF are tuned to obtain the best classification performances of three datasets. For the other

[1] https://github.com/ShuzhenZhang2019/SHDA

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projection transformation-based methods, the 5×5 weighted mean filter is applied to preprocess the hyperspectral data. The reduced dimensionality of the LDA method for the three datasets is set to be $N_c - 1$ (N_c is the number of land-cover classes in each data), and the reduced dimensionality d of the other projection transformation-based methods is fixed to be 30. The other parameters for LFDA, LPNPE, and SSDR are tuned to reach the best results for each data, while the other parameters for RLDE, SuperPCA, SuULDA, LCMR, and SBD are set to the default values suggested in [23], [37], [41], [36], and [46].

For all the compared projection transformation-based methods, the NN classifier is employed to obtain the classification results. For LCMR and SBD, since the extracted features lie on the Riemannian space, the suggested classifiers in [36] and [46] are, respectively, adopted. Five labeled pixels of each class are randomly selected for training samples and the rest for testing. All experiments are repeated ten times, and the overall classification accuracy (OA), average classification accuracy (AA), and the kappa coefficient (κ) are adopted to assess the performance of different methods.

C. Parameter Analysis

For the proposed SHDA method, the proportionality coefficient μ in formula (6) to computer the superpixel number is set to 0.04. The adjusting parameter ε in formula (8) to calculate the number of NN pixels in one 3-D superpixel is fixed to 50. The parameter pair (η, γ), which is used for constructing the scatter matrices, is set to $(0.6, 0.3)$. In this section, important parameters of the proposed SHDA method including the proportionality coefficient μ, the adjusting parameter ε, the regularization parameter η, the tradeoff parameter γ, and the number of reduced dimensionality d will be analyzed in detail. When the influence of one or two parameters is analyzed, the other parameters are fixed to the given default values.

First, the effect of the proportionality coefficient μ is discussed, which is utilized to compute the superpixel number. In experiments, the proportionality coefficient μ is ranged from 0.01 to 0.1 with step 0.01. Theoretically, too small μ will generate large subregions and may introduce some pixels from different classes in one superpixel, while too large μ will lead to small subregions and may not fully utilize pixel information in one homogeneous area. As can be seen from Fig. 2, for the three datasets of different scenes, the better OA values are obtained with various value ranges of μ, i.e., $\mu \in [0.03, 0.05]$ for the Indian Pines dataset, $\mu \in [0.04, 0.1]$ for the Pavia University dataset and $\mu \in [0.01, 0.04]$ for the Salinas dataset. Therefore, the intersection of the three ranges for the proportionality coefficient μ, i.e., $\mu = 0.04$, is selected for the remaining experiments.

Second, the effect of the adjusting parameter ε on the proposed SHDA method is analyzed, which is used to calculate the number of NN pixel p in SWMF. In the SHDA, the appropriate value of p can accelerate the SWMF process without decreasing the classification accuracy. According to formula (8), as the value of ε increases, the value of p decreases linearly. The variation tendencies of them are opposite. In experiments, the parameter ε is varied from 0 to 100 with a step of 10. The experimental results are shown in Fig. 3. The better classification results appear when $\varepsilon \in [30\,100]$ for Indian Pines dataset, $\varepsilon \in [20, 50]$ for Pavia University dataset and $\varepsilon \in [0, 60]$ for Salinas dataset. The intersection of the three ranges for ε is $[30, 50]$. Considering that selecting fewer neighbor pixels can reduce the computational amount of smoothing processing, the ε is fixed to 50 for the remaining experiments.

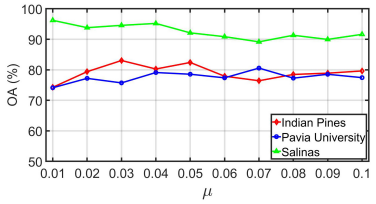


Fig. 2. Effect of the proportionality coefficient μ to computer the superpixel number on the proposed SHDA method for the three datasets.

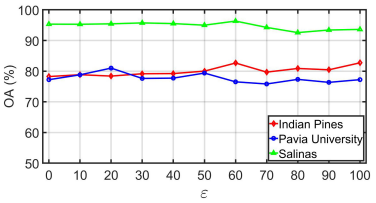


Fig. 3. Sensitivity analysis of the parameter ε to calculate the number of NN pixels on the proposed SHDA method for the three datasets.

Third, the sensitivity of the regularization parameter η and the tradeoff parameter γ are investigated. In experiments, η and γ are varied from 0 to 1 with step of 0.1. According to formulas (21) and (22), when $\gamma = 1$ and $\gamma = 0$, the SHDA method, respectively, reverts back to the SLDA method and the SNDA method. For the SNDA method, when $\eta = 0$, the SNDA method can be considered as the superpixel-level LDE method. Fig. 4 shows the OA of the proposed SHDA method as a function of η and γ. For the three datasets, the values of η and γ for gaining the better classification results exist the common ranges: $\eta \in [0.5, 0.8]$ and $\gamma \in [0.2, 0.4]$. Therefore, the mid-values of the two ranges, i.e., $\eta = 0.6$ and $\gamma = 0.3$, are applied for the subsequent experiments.

Finally, the influence of the reduced dimensionality d is analyzed. In experiments, the parameter d is varied from 2 to 50 with step of 1. Fig. 5 illustrates the variations of OA with the reduced dimensionality for the three datasets. Similar to most projection transformation-based methods [23], with the increase of the reduced dimensionality d, the OA values increase at first and then stay in a relatively stable trend around a high value. For the Indian Pines, Pavia University, and Salinas datasets, the better OA values are obtained when d are greater than 12, 10, and 8, respectively. Considering that, when the reduced dimensionality is set to 30, compared methods except LDA can achieve relatively stable and high OA values for the three datasets, the parameter $d = 30$ is selected for the remaining experiments.

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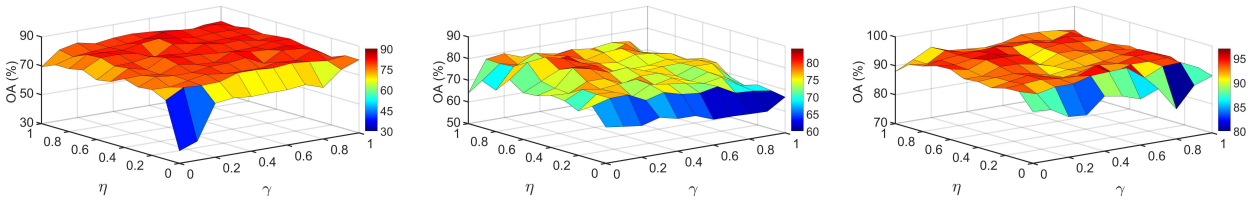


Fig. 4. Effect of the regularization parameter η and the tradeoff parameter γ on the proposed SHDA method. (a) Indian Pines dataset, (b) Pavia University dataset, and (c) Salinas dataset.

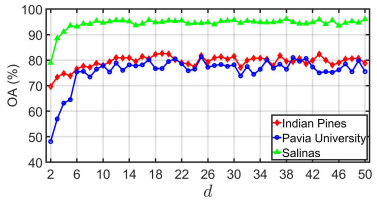


Fig. 5. Influence of the reduced dimensionality d on the proposed SHDA method for the three datasets.

TABLE II

CLASSIFICATION ACCURACY (IN PERCENT) OF THE ABLATION EXPERIMENT ON THE INDIAN PINES DATASET WITH FIVE TRAINING SAMPLES PER CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	Baseline	Baseline + SLDA	Baseline + SNDA	SHDA
1	97.80	98.05	98.05	**98.29**
2	51.92	**74.21**	54.41	70.43
3	67.19	**83.22**	73.82	73.08
4	72.33	**87.67**	81.25	86.12
5	77.36	72.57	77.03	**82.97**
6	78.57	92.39	90.98	**95.59**
7	**98.26**	97.39	95.65	96.96
8	95.56	**99.83**	88.50	96.96
9	**100**	**100**	**100**	**100**
10	74.07	**80.50**	76.06	71.37
11	56.30	65.50	57.86	**75.61**
12	61.96	**84.17**	79.23	82.60
13	99.50	**99.55**	**99.55**	99.50
14	84.13	85.85	86.60	**90.64**
15	63.42	75.59	76.19	**88.43**
16	**100**	**100**	**100**	98.86
OA	68.63	79.14	72.60	**80.87**
AA	79.90	87.28	83.45	**87.96**
κ	64.78	76.50	69.27	**78.45**

D. Module Ablation Analysis

In this section, to evaluate the effectiveness of different modules of the proposed SHDA method, the module ablation analysis is conducted. Specifically, the HSI processed by SWMF is first directly fed into NN classifier without any dimension-reduced operation, which is denoted as the baseline method. Then, the baseline method is combined with two main modules, i.e., SLDA and SNDA, to obtain the classification results.

Tables II–IV list the classification accuracy of three ablation approaches and the proposed SHDA method on the three datasets. The corresponding classification maps are shown in Figs. 6–8. As can be seen from these classification results, the baseline method almost performs the worst performance on the three datasets. This indicates that although SWMF can enhance the HSI quality, the processed HSI by SWMF still

TABLE III

CLASSIFICATION ACCURACY (IN PERCENT) OF THE ABLATION EXPERIMENT ON THE PAVIA UNIVERSITY DATASET WITH FIVE TRAINING SAMPLES PER CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	Baseline	Baseline + SLDA	Baseline + SNDA	SHDA
1	66.54	52.21	68.32	**76.65**
2	54.15	70.01	62.51	**75.12**
3	72.13	84.57	**87.17**	83.52
4	59.20	42.75	**72.75**	66.11
5	**97.28**	93.93	96.96	97.08
6	69.64	90.19	85.57	**95.47**
7	91.60	99.80	97.08	**100**
8	71.20	70.07	**77.76**	74.04
9	93.98	85.11	91.20	**96.34**
OA	63.99	70.39	72.16	**79.35**
AA	75.08	76.51	82.15	**84.93**
κ	55.73	63.13	65.61	**74.00**

TABLE IV

CLASSIFICATION ACCURACY (IN PERCENT) OF THE ABLATION EXPERIMENT ON THE SALINAS DATASET WITH FIVE TRAINING SAMPLES PER CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	Baseline	Baseline + SLDA	Baseline + SNDA	SHDA
1	**100**	**100**	**100**	**100**
2	99.79	99.79	99.79	**99.90**
3	96.05	**100**	95.46	**100**
4	**99.93**	**99.93**	**99.93**	**99.93**
5	99.33	91.44	**99.34**	99.33
6	99.60	**99.92**	99.86	**99.92**
7	95.86	99.78	89.45	**99.92**
8	69.35	72.67	80.98	**83.48**
9	95.35	**99.99**	99.07	99.78
10	81.95	96.41	92.93	**96.51**
11	99.76	**100**	**100**	**100**
12	**100**	**100**	**100**	**100**
13	97.80	97.96	97.98	**98.14**
14	**96.28**	93.21	94.00	93.48
15	86.06	82.54	**97.29**	94.82
16	97.48	**97.78**	94.48	97.35
OA	89.41	91.04	93.88	**95.32**
AA	94.66	95.71	96.28	**97.66**
κ	88.27	90.05	93.21	**94.81**

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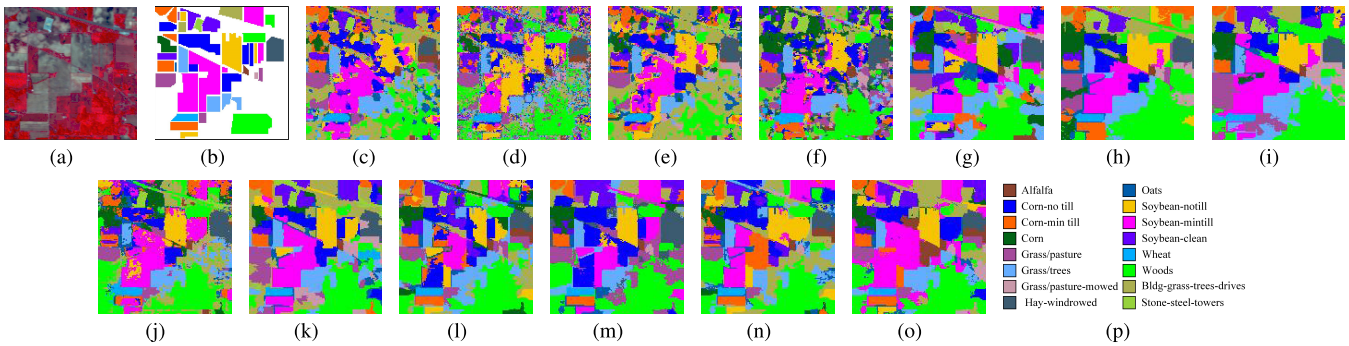


Fig. 6. Indian Pines dataset. (a) Pseudocolor map. (b) Ground truth. Classification maps obtained by different feature extraction methods (c) LDA [13], OA = 66.30%, (d) LFDA [19], OA = 64.80%, (e) RLDE [23], OA = 67.07%, (f) LPNPE [23], OA = 68.54%, (g) SuperPCA [37], OA = 64.04%, (h) SuULDA [41], OA = 78.76%, (i) SSDR [42], OA = 77.60%, (j) LCMR [46], OA = 72.68%, (k) SBD [36], OA = 76.47%, (l) SWMF (i.e., Baseline) [38], OA = 68.87%, (m) Baseline+SLDA, OA = 79.15%, (n) Baseline+SNDA OA = 72.65%, and (o) SHDA, OA = 80.81%. (p) Labels.

contains lots of redundant information and feature extraction is necessary. In addition, comparing the classification performances obtained by the other two ablation methods, for the Indian Pines dataset, the method combined with SLDA is superior to the method combined with SNDA. While, for the Pavia University and Salinas datasets, the method combined with SNDA outperforms the method combined with SLDA. It means that, for SLDA and SNDA modules, neither of them is intrinsically better than the other for the HSI classification. Meanwhile, the proposed SHDA method always achieves the best classification accuracy on the three datasets, which indicates that fusing the superpixel-level local and nonlocal information can improve the classification performance.

E. Comparison With Other Methods

In this section, extensive experiments are conducted to compare the proposed SHDA method with the classical and state-of-the-art feature extraction methods. For the three datasets used in experiments, five samples from each class are randomly selected to form the training set, and the remaining labeled samples are used for evaluation purposes. It means that only about 0.78%, 0.11%, and 0.15% of the labeled samples are used for training for the Indian Pines, Pavia University, and Salinas datasets, respectively. Hence, it is a very challenging task.

The quantitative results on the three datasets are listed in Tables V–VII and the corresponding classification maps are provided in Figs. 6–8. From these results, we can see that the traditional projection transformation methods, such as LDA and LFDA, show poor classification performances. In contrast, for the modified projection transformation methods, the RLDE method obtains better classification accuracy because it adopts a regularization strategy to ameliorate the small-sample-size problem. LPNPE employs spatial neighborhood information to learn the discriminant projections, which produces better classification performances on the Indian Pines dataset and Salinas dataset. However, for the Pavia University dataset, LPNPE performs worse. The reason is that LPNPE adopts the fixed window to capture local spatial–spectral information, which may be unsuitable for the Pavia University dataset with a rich texture structure. In addition, compared to the state-of-the-art superpixel-level projection transformation methods (i.e., SuperPCA, SuULDA, SSDR, and SHDA), the SuperPCA based on the PCA yields worse classification results than the others based on the discriminant analysis, which indicates that the discriminant analysis can generate more class-discernible features than the PCA. Moreover, for the state-of-the-art feature extraction methods, which are not based on projection transformation, LCMR and SBD gain high classification accuracy by extracting local spatial–spectral correlation features. In terms of quantitative results, the proposed SHDA method outperforms all of the compared methods. The classification maps of the three datasets obtained by the SHDA method are also the spatially smoothest and have the least mislabeled pixels. The reason is twofold. On one hand, the proposed SHDA method is a superpixel-level discriminant analysis, which can more reasonably utilize the spatial-structure information to learn the discriminant projections. On the other hand, in the proposed SHDA framework, both the local and nonlocal spatial–spectral correlation information between and among different superpixels is fully exploited and effectively fused, which produces more discriminant features for the HSI classification.

In addition, the computing times of the compared methods are reported in Table VIII. All the experiments are conducted by using MATLAB R2020b with an Intel(R) Core(TM) i7-9700 CPU at 3 GHz and 16 GB of RAM. As can be seen from Table VIII, for the Indian Pines dataset, the computing time of all the methods is competitive, which is within or around 10 s. However, for the Pavia University dataset and Salinas dataset with larger spatial size, the consuming-time gaps among different methods increase. In specific, the LCMR, SuperPCA, SuULDA, SSDR, and the proposed SHDA methods consume more time than the others. The reasons for the long-running times of these methods are different. For LCMR, it is time-consuming for each pixel to compute the local covariance matrix. For the superpixel-level methods, i.e., SuperPCA, SuULDA, SSDR, and SHDA, the processing of SWMF, which utilizes a superpixel-based weighted filter on each pixel to enhance the HSI quality, spends much computing time. Although among all the compared methods, the proposed SHDA method has no advantage in computing time, it achieves

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TABLE V

CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS ON INDIAN PINES DATASET WITH FIVE TRAINING SAMPLES PER CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	LDA	LFDA	RLDE	LPNPE	SuperPCA	SuULDA	SSDR	LCMR	SBD	SHDA
1	97.80	96.10	97.56	97.80	**100**	**100**	97.80	99.27	97.65	98.29
2	58.16	47.88	52.45	54.65	47.83	55.76	59.13	66.17	58.84	**70.43**
3	56.06	53.18	59.93	52.73	70.40	72.51	**77.43**	52.01	76.79	73.08
4	81.34	78.45	73.66	**94.87**	83.62	80.17	91.03	89.57	74.87	86.12
5	81.80	78.01	75.21	77.68	85.08	**85.50**	80.04	83.03	84.77	82.97
6	89.67	87.41	92.80	89.34	72.00	**98.23**	93.10	85.75	96.90	95.59
7	98.70	**100**	99.57	**100**	99.57	97.39	85.65	**100**	86.96	96.96
8	91.59	78.77	80.93	89.28	99.62	99.62	**99.81**	98.52	96.66	96.96
9	99.33	99.33	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
10	60.52	65.00	66.48	68.73	68.46	**84.98**	74.86	71.88	70.05	71.37
11	45.05	49.72	51.00	50.75	51.03	**75.79**	72.84	57.33	66.47	75.61
12	60.37	55.60	59.80	72.02	58.25	52.60	65.56	64.49	82.33	**82.60**
13	98.60	98.65	99.10	**99.55**	99.50	99.50	99.50	99.10	99.50	99.50
14	84.92	85.65	88.11	77.48	54.56	86.06	87.79	**94.40**	82.75	90.64
15	76.09	74.91	76.04	83.52	86.35	91.47	**94.33**	85.28	90.05	88.43
16	95.68	90.34	97.73	98.41	91.25	90.34	**100**	96.02	95.23	98.86
OA	66.04	63.64	67.19	67.50	63.77	78.15	78.11	72.95	76.14	**80.87**
AA	79.73	77.44	79.40	81.68	79.22	85.62	86.18	83.93	84.98	**87.96**
κ	62.04	60.40	63.29	63.81	59.85	75.35	75.38	69.66	73.15	**78.45**

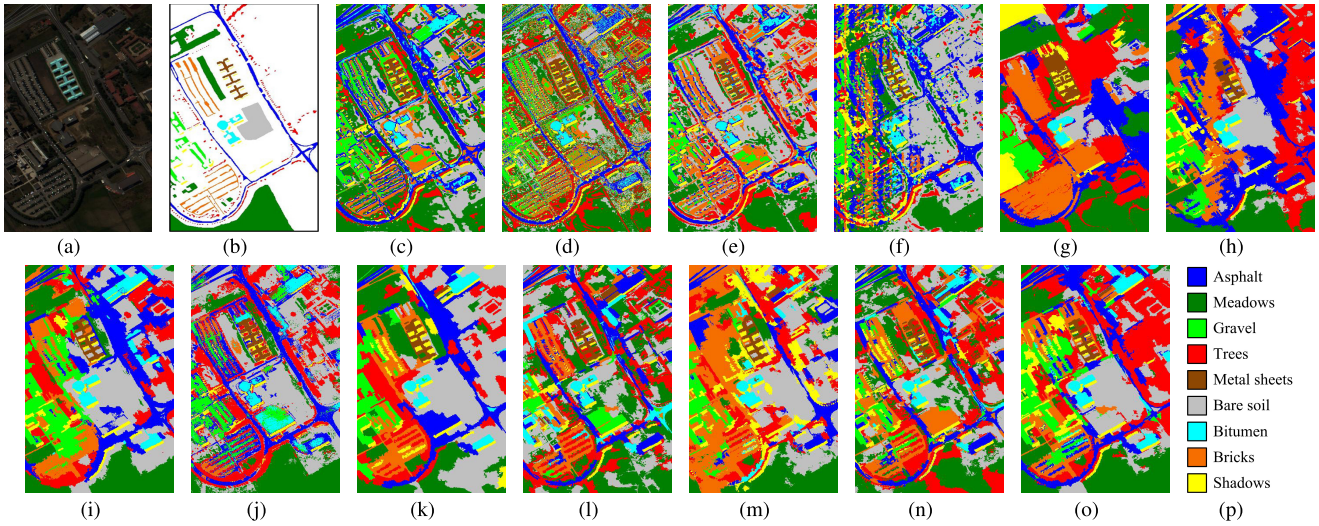


Fig. 7. Pavia University dataset. (a) Pseudocolor map. (b) Ground truth. Classification maps obtained by different feature extraction methods (c) LDA [13], OA = 75.30%, (d) LFDA [19], OA = 70.60%, (e) RLDE [23], OA = 77.13%, (f) LPNPE [23], OA = 67.36%, (g) SuperPCA [37], OA = 72.25%, (h) SuULDA [41], OA = 74.43%, (i) SSDR [42], OA = 77.08%, (j) LCMR [46], OA = 77.17%, (k) SBD [36], OA = 76.87%, (l) SWMF (i.e., Baseline) [38], OA = 63.79%, (m) Baseline + SLDA, OA = 70.38%, (n) Baseline+SNDA OA = 73.29%, and (o) SHDA, OA = 79.65%. (p) Labels.

TABLE VI

CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS ON PAVIA UNIVERSITY DATASET WITH FIVE TRAINING SAMPLES PER CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	LDA	LFDA	RLDE	LPNPE	SuperPCA	SuULDA	SSDR	LCMR	SBD	SHDA
1	**79.04**	76.35	73.58	56.34	49.07	61.18	75.24	71.84	59.74	76.65
2	68.15	62.78	70.78	64.11	73.36	73.97	75.34	75.82	**79.12**	75.12
3	75.44	69.44	74.21	74.34	86.30	**94.17**	77.75	74.42	88.67	83.52
4	86.36	86.56	89.15	68.41	46.09	52.29	63.95	**93.98**	53.92	66.11
5	99.49	**99.84**	99.72	96.01	93.32	94.48	96.88	95.70	95.75	97.08
6	80.60	72.51	82.96	87.64	94.72	**96.71**	82.48	84.17	87.94	95.47
7	90.97	91.02	94.42	88.84	96.30	**100**	99.98	86.27	91.64	**100**
8	67.35	67.05	74.91	50.93	**83.99**	70.65	74.33	61.21	77.01	74.04
9	99.49	99.63	**99.81**	99.54	49.41	88.67	94.07	91.36	75.15	96.34
OA	75.28	71.28	76.76	67.89	72.51	75.59	77.23	77.45	76.46	**79.35**
AA	83.00	80.58	84.39	76.24	74.73	81.35	82.22	81.64	78.77	**84.93**
κ	69.13	64.23	70.78	60.45	65.46	69.36	71.13	71.46	70.07	**74.00**

a better balance between efficiency and effectiveness of the algorithm.

To further verify the proposed SHDA method, we have also investigated the effect of the number of training samples

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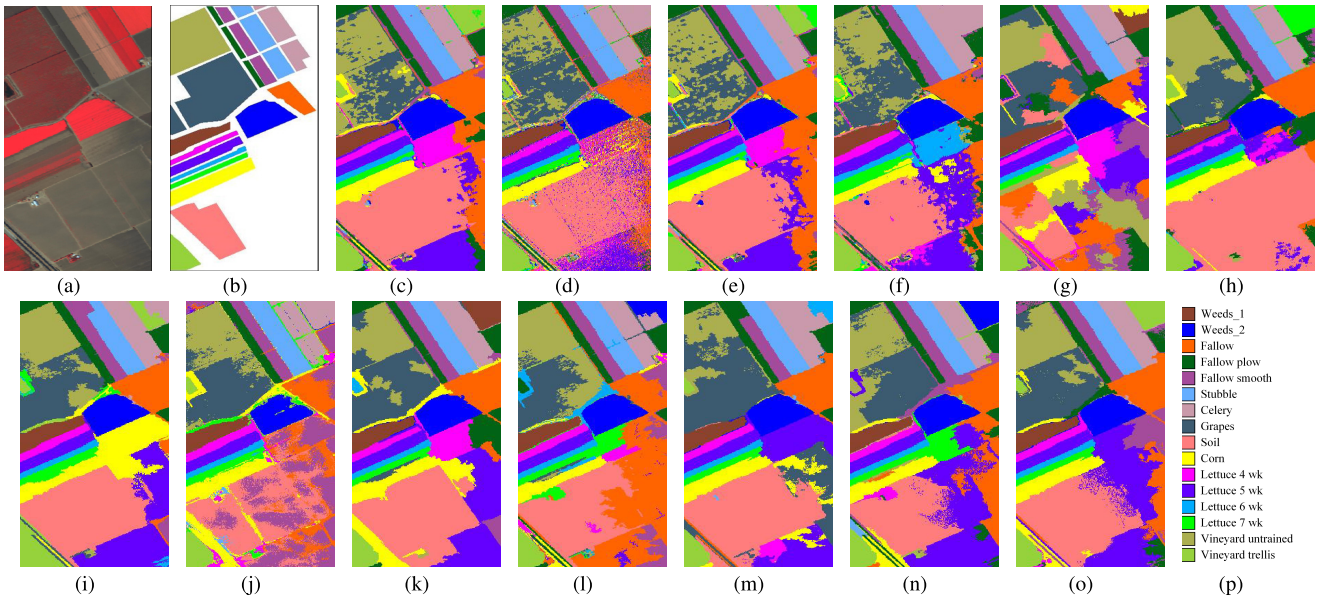


Fig. 8. Salinas dataset. (a) Pseudocolor map. (b) Ground truth. Classification maps obtained by different feature extraction methods (c) LDA [13], OA = 88.70%, (d) LFDA [19], OA = 89.21%, (e) RLDE [23], OA = 89.64%, (f) LPNPE [23], OA = 89.57%, (g) SuperPCA [37], OA = 79.34%, (h) SuULDA [41], OA = 93.98%, (i) SSDR [42], OA = 92.96%, (j) LCMR [46], OA = 91.64%, (k) SBD [36], OA=93.47%, (l) SWMF (i.e., Baseline) [38], OA = 89.48%, (m) Baseline+SLDA, OA = 91.32%, (n) Baseline + SNDA OA = 93.19%, and (o) SHDA, OA = 95.80%. (p) Labels.

TABLE VII

CLASSIFICATION ACCURACY (IN PERCENT) OF DIFFERENT METHODS ON SALINAS DATASET WITH FIVE TRAINING SAMPLES PER CLASS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	LDA	LFDA	RLDE	LPNPE	SuperPCA	SuULDA	SSDR	LCMR	SBD	SHDA
1	98.72	99.89	99.86	96.43	88.89	**100**	**100**	99.89	**100**	**100**
2	99.80	99.54	98.38	**100**	94.46	99.79	99.83	83.16	99.96	99.90
3	99.42	99.11	97.66	99.86	85.91	**100**	98.74	97.95	**100**	**100**
4	98.73	99.14	99.22	98.06	87.12	93.38	78.73	98.01	**99.93**	**99.93**
5	96.49	98.50	95.88	96.65	91.61	92.68	85.48	95.87	99.27	**99.33**
6	99.37	99.84	99.47	99.90	98.60	99.67	**99.92**	98.36	**99.92**	**99.92**
7	99.54	99.72	99.73	**99.93**	76.98	99.31	99.27	96.23	99.48	**99.93**
8	65.02	68.44	67.45	67.83	66.50	77.67	76.57	83.10	79.86	**83.48**
9	99.96	99.97	99.99	98.89	69.00	99.98	**100**	91.22	99.80	99.78
10	87.47	91.55	84.36	90.26	85.72	93.92	96.17	90.06	**97.28**	96.51
11	95.23	96.43	96.92	98.81	93.73	91.83	94.17	99.96	97.29	**100**
12	99.99	**100**	99.95	96.15	80.48	93.52	99.52	92.36	**100**	**100**
13	99.09	99.24	**99.48**	98.34	98.32	98.23	97.80	98.19	97.80	98.14
14	94.06	93.29	96.50	88.69	90.80	95.21	96.21	93.70	**96.54**	93.48
15	76.68	77.91	77.11	85.15	65.35	**96.18**	93.43	87.07	85.73	94.82
16	96.57	97.49	97.44	98.02	99.06	**99.15**	97.78	92.09	97.99	97.35
OA	88.11	89.44	88.48	89.74	79.20	93.30	92.31	90.58	93.39	**95.32**
AA	94.13	95.00	94.34	94.56	85.78	95.66	94.60	93.58	96.93	**97.66**
κ	86.80	88.29	87.22	88.63	77.14	92.57	91.48	89.53	92.66	**94.81**

TABLE VIII

COMPUTING TIME (SECONDS) FOR THE CLASSIFICATION OF THREE DATASETS BY THE LDA [13], LFDA [19], RLDE [23], LPNPE [23], SUPERPCA [37], SUULDA [41], SSDR [42], LCMR [46], SBD [36], AND SHDA METHODS

Data sets	LDA	LFDA	RLDE	LPNPE	SuperPCA	SuULDA	SSDR	LCMR	SBD	SHDA
Indian Pines	1.31	0.82	1.29	1.37	4.84	5.57	7.35	6.52	11.71	5.77
Pavia University	9.33	2.52	9.71	8.21	191.01	54.50	66.48	58.47	7.78	69.51
Salinas	8.02	4.68	8.37	8.35	196.63	38.78	164.81	33.93	15.06	101.55

for compared methods on three datasets. In experiments, we randomly select different labeled samples from each class to be the training set and the rest to be the testing set. The number of the selected labeled samples per class is varied from 5 to 50 with the step of 5. Here, for the Indian Pines dataset, since the labeled sample number of three categories (alfalfa, grass-pasture-mowed, and oats) is less than 50, half of the labeled samples of these categories are chosen for the

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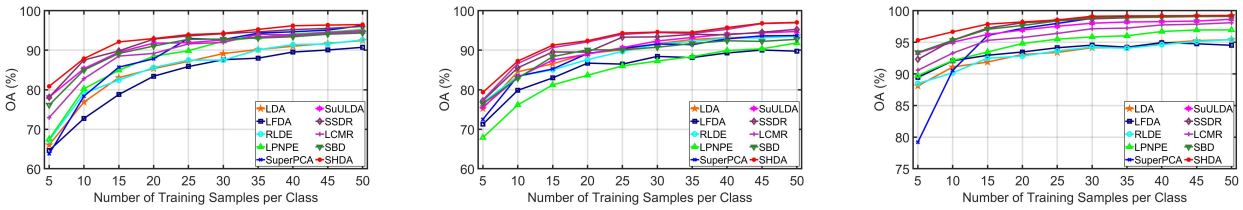


Fig. 9. Effect of the number of training samples on the LDA [13], LFDA [19], RLDE [23], LPNPE [23], SuperPCA [37], SuULDA [41], SSDR [42], LCMR [46], SBD [36], and SHDA methods for (a) Indian Pines dataset, (b) Pavia University dataset, and (c) Salinas dataset.

training. As shown in Fig. 9, with the increase in the number of training samples, the classification performances for all the considered feature extraction methods are generally improved. Most importantly, the proposed SHDA method consistently provides better performances than all other contrastive methods. These results further demonstrate that, by fusing the superpixel-level local and nonlocal information, the extracted features obtained by SHDA have more discriminative capability for HSI classification.

V. Conclusion

In this article, a novel SHDA method has been proposed for HSI feature extraction. In this method, to extract the strong class-discriminative features, the complex spatial–spectral information of HSI is fully explored from the superpixel-level local and nonlocal perspectives. Specifically, from the superpixel-level local perspective, the SLDA module skilly employs spectral similarity information within superpixel and differential information among adjacent superpixels into the LDA scheme. From the superpixel-level nonlocal perspective, the SNDA module utilizes contextual relations of superpixels to construct a graph embedding scheme. Then, the SLDA and SNDA modules are effectively fused in the SHDA method to learn the projection transformation. By projection transformation, the high-dimensional data projects into a low-dimensional feature space with great class separability. Contrast experiments on three real-world HSI datasets demonstrate that the proposed SHDA method outperforms the existing classic and state-of-the-art feature extraction methods. In future works, we would integrate the multiscale spatial–spectral information into the SHDA framework to learn the low-dimensional feature with more powerfully discriminative performance.

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